

Bachelor in Computer Applications

Course Title: **Digital Logics**

Code No: Semester: **1**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

1. Design a combinational circuit with four inputs and one output. The output is equal to 1 when (i) all the inputs are equal to 1 or (ii) none of the inputs are equal to 1 or (iii) an odd number of inputs are equal to 1.
2. Implement the following function: $F = \sum(0,1,3,4,5,8,9,10,15)$ using (i) Decoder (ii) Multiplexer (iii) PLA
3. What do you mean by race condition in JK Flip Flop and mention the methods to overcome it? Explain Master Slave Flip-flop using JK flip flop with logic circuit, truth table and timing diagram.
4. Convert (1011.110) into decimal and hexadecimal.
5. Convert (51966.57)₁₀ into octal and hexadecimal number system.
6. Subtract (111000.110)₂ — (110100.101)₂ using both 2's and 1's complement.
7. State and prove De-Morgan's Theorems.
8. Define Half-subtractor with truth table and logic diagram.
9. What is decoder? Implement 8 x 1 MUX using 4 x 1 MUX.
10. What is clocked RS flip-flop? Explain with logic diagram and characteristic table.
11. Design Mod-3 synchronous counter.
12. Draw a Parallel-In Parallel-Out Shift register and explain it.